IP ROUTING AND ROUTING PROTOCOLS

Course: Diploma in Cyber Security
Topic: IP Routing and Routing Protocols
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Introduction

IP routing is the process of choosing a path for packets to travel between different networks. Routers use routing tables to make these decisions. In this assignment, I have covered a simple routing example, a sample routing table, static and dynamic routing, and an OSPF example.

Task 1: Three Devices Through a Router

- Phone: 192.168.1.10
- Smart TV: 192.168.1.20
- Friend's Laptop: 192.168.2.10
- Router LAN interface: 192.168.1.1
- Router second interface: 192.168.2.1

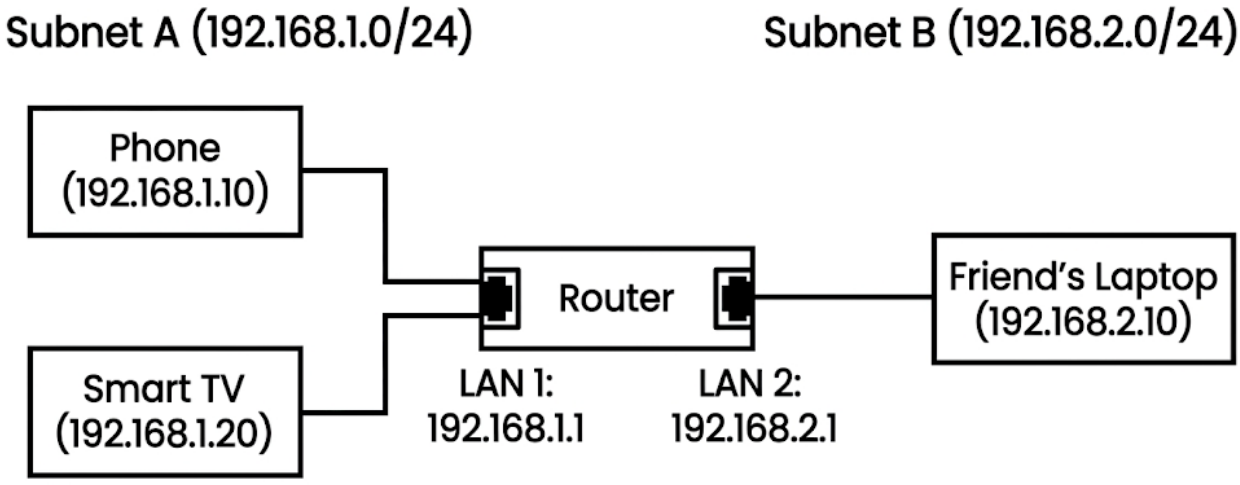


Figure 1: Devices Connected Through a Router

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Task 2: Manual IP Routing Example

Example: Laptop A = 192.168.1.10/24 and Friend's PC = 192.168.2.20/24. Router R1 has 192.168.1.1 on LAN 1 and 192.168.2.1 on LAN 2.

1. Laptop A sees that 192.168.2.20 is outside its local 192.168.1.0/24 network, so it sends the packet to its default gateway 192.168.1.1.

2. Router R1 checks its routing table and finds a route for 192.168.2.0/24 through its second interface.

3. R1 forwards the packet through that interface to the 192.168.2.0/24 network.

4. The destination PC receives the packet and the reply follows the routing information back.

Sample routing table entry: 192.168.2.0/24 -> directly connected -> Interface G0/1

Task 3: Fictional Zomato-Style Delivery Network

Branch	Network	Mask	Next Hop	Interface
Ahmedabad	10.10.1.0	255.255.255.0	Directly connected	G0/0
Vadodara	10.10.2.0	255.255.255.0	10.10.0.2	G0/1
Surat	10.10.3.0	255.255.255.0	10.10.0.6	G0/2

The next-hop addresses are example values for a fictional network.

Task 4: Static vs Dynamic Routing

Type	Pros (3)	Cons (3)	Example

Static Routing	**Simple; predictable; no routing-protocol traffic.**	**Manual work; does not automatically react to failures; harder to manage at large scale.**	**Small office or fixed branch network.**
Dynamic Routing (RIP/OSPF)	**Automatically learns routes; can react to changes; useful for larger networks.**	**More setup; uses CPU/memory; configuration is more complex.**	**Large delivery-service network with multiple branches.**

Task 5: Sample OSPF Configuration

Example Cisco IOS configuration for two routers:

Router 1
enable
configure terminal
router ospf 1
network 10.0.1.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.255 area 0

Router 2
enable
configure terminal
router ospf 1

network 10.0.2.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.255 area 0

- enable = enters privileged mode.
- configure terminal = enters global configuration mode.
- router ospf 1 = starts OSPF process 1.
- network ... area 0 = tells OSPF which network to include and places it in area 0.

Task 3: Cisco 2-Router OSPF Configuration Topology

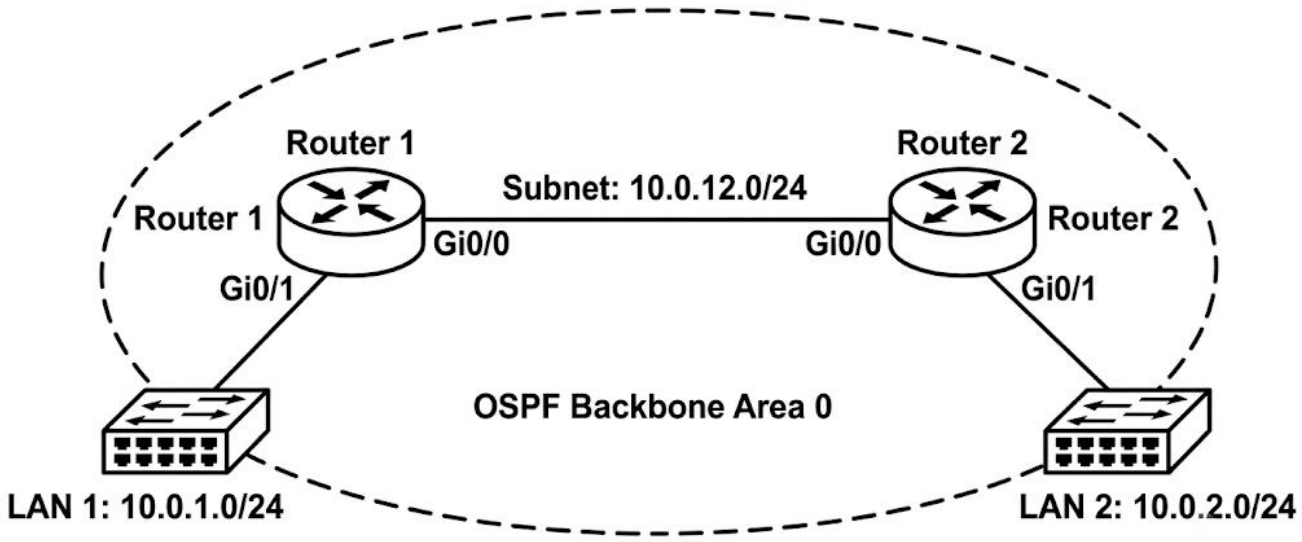


Figure: Cisco 2-Router OSPF Configuration Topology

Conclusion

I learned that routers use routing tables to choose paths for packets. Static routes are simple for small fixed networks, while dynamic protocols such as OSPF are useful when networks are larger or routes can change.

Quick Revision

- Router = forwards packets between networks.
- Routing table = information used to choose a path.
- Static routing = manually configured.
- Dynamic routing = routes are learned and updated automatically.
- OSPF = dynamic link-state routing protocol.